A QUASI-COMPACT QUASI-SEPARATED SCHEME WITH NO SEPARATED QUASI-COMPACT DENSE OPEN

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The argument's final contradiction is the author's own 'by the first paragraph of this section'; the first paragraph is included as the first context span.

Editorial note. The label lemma-no-dense-separated-quasi-compact-open-in-qcqs names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 2249-2274 (delivered as L48311): no quasi-compact open dense in S - {s}, and the T/U complement statement, including the isomorphism (T, t) = (S, s) sending x to z_1 and z_i to z_{i+1}
- Context span 2293-2312: X is the glueing of two copies of T along the affine open U; pi: X -> T maps U_i isomorphically to T and U_1 cap U_2 isomorphically to U; X is quasi-separated (Schemes, characterize-quasi-separated) and quasi-compact; for the closed points x_1 in U_1, x_2 in U_2 over t and the generic point etatilde in U_1 cap U_2 over eta, we have etatilde leads to x_1 and etatilde leads to x_2; since pi|_W: W -> T is separated, W cannot contain both x_1 and x_2 (valuative criterion, Schemes, valuative-criterion-separatedness); say x_1 not in W; then W cap U_1 is a quasi-compact (X quasi-separated) dense open of U_1 not containing x_1; via (T, t) = (S, s) this contradicts the first context span

Source reference. Original lemma 01KO: sources/stacks-chow/tags.txt, source lines 2057–2057. Prerequisite pointer invoked at 2353; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01L0: sources/stacks-chow/tags.txt, source lines 2069–2069. Prerequisite pointer invoked at 2363; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Let k be a field and let $A = k[z_1, z_2, z_3, \ldots]/I$ where I is the ideal generated by all pairwise products $z_i z_j$, $i \neq j$, $i, j \in \mathbf{N}$. Set $S = \mathrm{Spec}(A)$. Let $s \in S$ be the closed point corresponding to the maximal ideal (z_i). We claim there is no quasi-compact open $V \subset S \setminus \{s\}$ which is dense in $S \setminus \{s\}$. Note that $S \setminus \{s\} = \bigcup D(z_i)$. Each $D(z_i)$ is open and irreducible with generic point η_i. We conclude that $\eta_i \in V$ for all i. However, a principal affine open of $S \setminus \{s\}$ is of the form $D(f)$ where

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$f \in (z_1, z_2, \ldots)$. Then $f \in (z_1, \ldots, z_n)$ for some n and we see that $D(f)$ contains only finitely many of the points η_i. Thus V cannot be quasi-compact.

Let k be a field and let $B = k[x, z_1, z_2, z_3, \ldots]/J$ where J is the ideal generated by the products xz_i, $i \in \mathbf{N}$ and by all pairwise products $z_i z_j$, $i \neq j$, $i, j \in \mathbf{N}$. Set $T = \mathrm{Spec}(B)$. Consider the principal open $U = D(x)$. We claim there is no quasi-compact open $V \subset T$ such that $V \cap U = \emptyset$ and $V \cup U$ is dense in T. Let $t \in T$ be the closed point corresponding to the maximal ideal (x, z_i). The closure of U in T is $\overline{U} = U \cup \{t\}$. Hence $V \subset \bigcup_i D(z_i)$ is a quasi-compact open. By the arguments of the previous paragraph we see that V cannot be dense in $\bigcup D(z_i)$.

Let X be the glueing of two copies of the affine scheme T (see above) along the affine open U. Thus there is a morphism $\pi : X \to T$ and $X = U_1 \cup U_2$ such that π maps U_i isomorphically to T and $U_1 \cap U_2$ isomorphically to U. Note that X is quasi-separated (by Schemes, Lemma 01KO) and quasi-compact. We claim there does not exist a separated, dense, quasi-compact open $W \subset X$. Namely, consider the two closed points $x_1 \in U_1$, $x_2 \in U_2$ mapping to the closed point $t \in T$ introduced above. Let $\tilde{\eta} \in U_1 \cap U_2$ be the generic point mapping to the (unique) generic point η of U. Note that $\tilde{\eta} \rightsquigarrow x_1$ and $\tilde{\eta} \rightsquigarrow x_2$ lying over the specialization $\eta \rightsquigarrow t$. Since $\pi|_W : W \to T$ is separated we conclude that we cannot have both x_1 and $x_2 \in W$ (by the valuative criterion of separatedness Schemes, Lemma 01L0). Say $x_1 \notin W$. Then $W \cap U_1$ is a quasi-compact (as X is quasi-separated) dense open of U_1 which does not contain x_1. Now observe that there exists an isomorphism $(T, t) \cong (S, s)$ of schemes (by sending x to z_1 and z_i to z_{i+1}). Hence by the first paragraph of this section we arrive at a contradiction.

Lemma 0.1. *There exists a quasi-compact and quasi-separated scheme X which does not contain a separated quasi-compact dense open.*

Proof. See discussion above. □